

Rigidity of the known five-dimensional kissing configurations and a proof of a conjecture of Cohn and Rajagopal

Stephan Mitterer*
s.mitterer@online360.at

September 2026

Abstract

We prove that the four known 40-point kissing configurations in $\mathbb{R}^5$ — the D_5 configuration, Leech’s L_5 , Szöllősi’s Q_5 , and Cohn–Rajagopal’s R_5 — are infinitesimally jammed, hence jammed; for Q_5 this answers a question of Szöllősi, and for D_5 , L_5 it reproduces results of Cohn, Jiao, Kumar, and Torquato. We certify in exact arithmetic that no 41st point can be adjoined to any of the four. We then prove Conjecture 3.1 of Cohn and Rajagopal: no six-dimensional kissing configuration with Q_5 or R_5 as a cross section contains 72 or more points; in fact at most 70. The proof combines a reduction lemma (any such cross section is central and the configuration splits into two independent layers), a norm-coupling inequality showing each layer projects to a spherical code in S^4 with inner products at most $1/5$, the sharp linear-programming bound $A(5, 1/5) = 16$ with its equality analysis à la Delsarte–Goethals–Seidel [10], and the exact computation that the deep-hole graphs of Q_5 and R_5 have clique number 12 (versus 16 for D_5, L_5). We also construct explicit 64-point six-dimensional kissing configurations with Q_5 and R_5 cross sections. All certificates are exact data (rational, or in $\mathbb{Q}(\sqrt{3})$ for the 64-point extensions), verified by independent computations, and accompany the paper.

1 Introduction

The kissing number $\kappa(5)$ satisfies $40 \leq \kappa(5) \leq 44$ [1, 5], and 40 is widely conjectured optimal. For decades, two non-isometric 40-point kissing configurations in $\mathbb{R}^5$ were known — the minimal vectors of the root lattice D_5 , and a non-lattice configuration of Leech [2] — and both were conjectured to be optimal; it was believed that no others existed [3, 4]. Szöllősi [6] found a third configuration Q_5 , refuting that belief, and Cohn and Rajagopal [7] subsequently found a fourth, R_5 . Szöllősi explicitly raised the rigidity of Q_5 as a question; the question matters because a flexible optimal configuration would open a deform-and-augment route towards $\kappa(5) \geq 41$.

We work at squared norm 2: following the usage of [7], a *kissing configuration* is a finite $X \subset \mathbb{R}^d$ with $\langle x, x \rangle = 2$ and $\langle x, y \rangle \leq 1$ for $x \neq y$ (no size restriction is implied; the term

*Computer assistance. All code, certificates and the first draft of this note were produced with Claude Fable 5 (Anthropic) under the author’s direction (with Claude Opus 5 and Claude Sonnet 5 as subagents for the verifier implementations and literature retrieval, respectively). The results were then independently re-verified from the committed code in fresh sessions with Claude Fable 5.1, Gemini Pro and MiniMax M3, and by the exact-arithmetic checks described in Section 5. The author is responsible for this publication, the repository and this text; the mathematics was developed by the systems named above and verified as described in Section 5.

refers to the touching condition, not to maximality); *contacts* are pairs with $\langle x, y \rangle = 1$. All four configurations have purely rational coordinates in this normalization [7, Table 2.2]; each has exactly 240 contacts forming a 12-regular graph.

Theorem 1. *Each of D_5 , L_5 , Q_5 , R_5 is infinitesimally jammed, hence jammed. The results for Q_5 and R_5 are new; those for D_5, L_5 reproduce [4].*

Theorem 2. *For each $X \in \{D_5, L_5, Q_5, R_5\}$, the polytope $P_X = \{w \in \mathbb{R}^5 : \langle x, w \rangle \leq 1 \ \forall x \in X\}$ is compact with $\max_{w \in P_X} |w|^2 = 5/4$, attained exactly at 32 vertices (the deep holes); consequently no 41st point can be adjoined to X .*

Theorem 3 (Conjecture 3.1 of [7]). *No six-dimensional kissing configuration containing a similarity copy of Q_5 or R_5 has more than 70 points. In particular none reaches 72.*

We emphasize what is *not* claimed: Theorems 1 and 2 are statements about the known configurations and imply nothing about hypothetical unrelated 41-point codes. Theorem 3 says nothing about $\kappa(6)$ itself: the E_6 configuration gives 72 kissing spheres in dimension six, while the exact value of $\kappa(6)$ remains unknown, the current bounds being $72 \leq \kappa(6) \leq 77$ [8]; our theorem only bounds configurations that contain Q_5 or R_5 . The bound 70 is likely not sharp (we construct 64-point extensions and conjecture 64 is optimal).

2 Jamming

Let $X = \{x_1, \dots, x_{40}\}$ be one of the four configurations. An *infinitesimal flex* is $(v_1, \dots, v_{40})$, $v_i \in \mathbb{R}^5$, with $\langle x_i, v_i \rangle = 0$ for all i and $g_{ij}(v) := \langle x_i, v_j \rangle + \langle x_j, v_i \rangle \leq 0$ for every contact $\{i, j\}$. Trivial flexes are $v_i = Ax_i$ with A antisymmetric; they form a 10-dimensional space since X spans $\mathbb{R}^5$. X is *infinitesimally jammed* if every flex is trivial; infinitesimally jammed implies jammed [4, Thm. 2.2] (Connelly; Roth–Whiteley).

Proof of Theorem 1. Two exact certificates per configuration.

(i) *A strictly positive self-stress:* weights $\omega_e > 0$ on the 240 contacts with $s_i := \sum_{j:\{i,j\} \in E} \omega_{ij} x_j$ parallel to x_i for every i . For D_5 and L_5 the uniform stress $\omega \equiv 1$ works, with $s_i = 6x_i$. For Q_5 the uniform stress provably fails; a valid stress takes the two values $20/21$ (on 210 contacts) and $4/3$ (on 30), with $s_i = \lambda_i x_i$, $\lambda_i \in \{40/7, 44/7\}$. For R_5 the values are $200/207$ ($\times 210$), $280/207$ ($\times 18$), $220/207$ ($\times 12$), with $\lambda_i \in \{400/69, 440/69, 140/23\}$. Verification is a finite exact computation. For any flex v ,

$$\sum_e \omega_e g_e(v) = \sum_i \langle s_i, v_i \rangle = \sum_i \lambda_i \langle x_i, v_i \rangle = 0,$$

and since every $\omega_e > 0$ and every $g_e(v) \leq 0$, all $g_e(v) = 0$: every flex preserves every contact to first order.

(ii) *The equality kernel is trivial:* the stacked linear system of the 40 tangency equations and 240 contact equalities in the 200 unknowns has rank exactly 190 over $\mathbb{Q}$ (computed by fraction-free Bareiss elimination, and independently via a mod- p argument: $\text{rank}_{\mathbb{F}_p} \leq \text{rank}_{\mathbb{Q}}$ together with the 10-dimensional space of rotation flexes in the kernel). Hence the kernel is exactly the trivial flexes, and X is infinitesimally jammed. $\square$

3 Deep holes: no 41st point

Proof of Theorem 2. $\sum_i x_i = 0$ exactly and X spans $\mathbb{R}^5$, so the origin is interior to $\text{conv}(X)$ and P_X is compact. Every vertex of P_X is the unique solution of $\langle x_i, w \rangle = 1$ for five linearly independent constraints; enumerating all $\binom{40}{5} = 658,008$ subsets in exact integer arithmetic yields the complete vertex lists: 42/50/92/100 vertices for $D_5/L_5/Q_5/R_5$, with maximal squared norm $5/4$ attained at exactly 32 vertices in each case (for D_5 these are the deep holes of [7, §3]; for Q_5, R_5 we confirm exactly the reflection structure described there). Since $|\cdot|^2$ is strictly convex, its maximum over P_X is attained only at vertices; a 41st point x would satisfy $x \in P_X$ and $|x|^2 = 2 > 5/4$. $\square$

For later use, define the *hole graph* of X : vertices the 32 deep holes, edges the pairs with $\langle w, w' \rangle \leq 1/4$ (the maximum inner product between distinct holes is $23/20$ for Q_5, R_5 and $3/4$ for D_5, L_5). In the normalization of [7, §3] the holes lie at squared norm 2 and the threshold is $2/5$; our holes and threshold are the same up to the factor $8/5$, and for unit vectors the threshold is the $1/5$ of Lemmas 7–8.

Lemma 4. *The clique number of the hole graph is 16 for D_5 and L_5 , and 12 for Q_5 and R_5 .*

Proof. Exact computation (branch-and-bound over the 32-vertex graphs; verified by two algorithms: a library Bron–Kerbosch enumeration and an own branch-and-bound). For D_5 the 16-cliques are the half-families $S_{\text{even}}, S_{\text{odd}}$ of [7, §3]. $\square$

4 Six-dimensional extensions

Lemma 5 (Reduction). *Let $Y \subset \mathbb{R}^6$ be a kissing configuration (squared norm 2) containing a similarity copy Z of $X \in \{Q_5, R_5\}$. Then:*

1. *Z is central, at ambient scale, and $Z = Y \cap H$ for the hyperplane H it spans; we may take $H = \{y_6 = 0\}$ and $Z = X \times \{0\}$.*
2. *Every other point of Y has the form $y = (u, h)$ with $u \in P_X$, $h^2 \geq 3/4$.*
3. *Any two such points with $h > 0 > h'$ are automatically compatible.*
4. *Consequently $|Y| = 40 + |Y^+| + |Y^-|$, where $Y^\pm$ are the layers $\{h \gtrless 0\}$, and each layer is a cap set: points $y_i = (u_i, h_i)$, $u_i \in P_X$, $h_i = \sqrt{2 - |u_i|^2}$, pairwise $\langle u_i, u_j \rangle + h_i h_j \leq 1$.*

Proof. (1) X has affine rank 5, so Z spans a hyperplane $H = \{\langle y, n \rangle = c\}$, $|n| = 1$; all points of Z lie on the ambient sphere S . Since X affinely spans $\mathbb{R}^5$ its circumsphere is unique, and the similarity carries it to the unique circumsphere of Z inside H , namely $H \cap S$ with center cn and squared radius $2 - c^2$; hence $Z = cn + \lambda \cdot (\text{orthogonal image of } X)$ with $\lambda^2 \cdot 2 = 2 - c^2$. X has 240 contact pairs at inner product exactly 1; their images satisfy $\langle y, y' \rangle = \lambda^2 \cdot 1 + c^2 = 1 + c^2/2$, which respects the kissing constraint only if $c = 0$. A point $(u, 0) \in Y \setminus Z$ would satisfy $u \in P_X$, hence $|u|^2 \leq 5/4 < 2$ by Theorem 2 — impossible for a point of squared norm 2. (2) Compatibility with every $(x, 0)$ gives $u \in P_X$; $|u|^2 \leq 5/4$ gives $h^2 = 2 - |u|^2 \geq 3/4$. (3) $\langle y, y' \rangle = \langle u, u' \rangle + hh' \leq |u||u'| + \sqrt{3/4}\sqrt{3/4} \leq 5/4 - 3/4 = 1/2 \leq 1$. (4) Immediate from (1)–(3); conversely any cap set and its mirror assemble with $X \times \{0\}$ into a valid configuration. $\square$

Theorem 6. *There exist 64-point six-dimensional kissing configurations with Q_5 (resp. R_5) as central cross section: the cross-section together with a 12-point hole clique placed at heights $\pm\sqrt{3}/2$. (Exact certificates accompany the paper.)*

We now bound cap sets. Write $M(X)$ for the maximum size of a cap set; by Lemma 5, Theorem 3 is equivalent to $M(Q_5), M(R_5) \leq 15$.

Lemma 7 (Norm coupling). *Let $y = (u, h)$, $y' = (u', h')$ be distinct compatible cap points with $u, u' \neq 0$. Then $\langle \hat{u}, \hat{u}' \rangle \leq 1/5$ for the unit vectors $\hat{u} = u/|u|$, $\hat{u}' = u'/|u'|$, with equality only if $|u|^2 = |u'|^2 = 5/4$.*

Proof. Compatibility gives $\langle \hat{u}, \hat{u}' \rangle \leq (1 - hh')/(|u||u'|)$. With $a = h$, $b = h'$, $|u| = \sqrt{2 - a^2}$, $|u'| = \sqrt{2 - b^2}$, and $a, b \in [\sqrt{3}/2, \sqrt{2}]$, the claim reduces to $G(a, b) := \sqrt{(2 - a^2)(2 - b^2)} - 5(1 - ab) \geq 0$. If $ab \geq 1$ this is trivial. If $ab < 1$, both sides of $\sqrt{(2 - a^2)(2 - b^2)} \geq 5(1 - ab)$ are nonnegative, and squaring gives the equivalent

$$2(a^2 + b^2) + 24(ab)^2 - 50ab + 21 \leq 0.$$

Set $x = ab \in [3/4, 1)$ and write $b = x/a$; the constraints $a, b \geq \sqrt{3}/2$ confine a to $[\sqrt{3}/2, 2x/\sqrt{3}]$. The function $\varphi(a) = a^2 + x^2/a^2$ is strictly convex for $a > 0$ ($\varphi''(a) = 2 + 6x^2/a^4 > 0$), so its maximum over this interval is attained at an endpoint, and by the symmetry $a \leftrightarrow x/a$ both endpoints give the same value $\varphi(\sqrt{3}/2) = 3/4 + 4x^2/3$. Hence the left side is at most $(80/3)x^2 - 50x + 45/2$, a quadratic with roots exactly $x = 3/4$ and $x = 9/8$, so it is ≤ 0 on $[3/4, 9/8] \supset [3/4, 1)$. Equality forces $x = 3/4$ and $a = b = \sqrt{3}/2$, i.e. $|u|^2 = |u'|^2 = 5/4$; tracing back, equality in the lemma also requires $\langle u, u' \rangle = 1 - hh'$. In angular variables $|u| = \sqrt{2} \sin \theta$, $h = \sqrt{2} \cos \theta$ the claim reads $10 \cos \theta \cos \theta' + 2 \sin \theta \sin \theta' = 6 \cos(\theta - \theta') + 4 \cos(\theta + \theta') \geq 5$ on $(0, \theta^*]^2$, $\sin^2 \theta^* = 5/8$: for fixed $\theta + \theta'$ the left side decreases in $|\theta - \theta'|$, so the minimum is on the edge $\theta' = \theta^*$, where $g(\theta) = 6 \cos(\theta - \theta^*) + 4 \cos(\theta + \theta^*)$ is concave with $g(\theta^*) = 5 < g(0)$. $\square$

Two degenerate remarks: the pole point $(0, \sqrt{2})$ is incompatible with every other cap point ($g = \sqrt{2}h' \geq \sqrt{3}/2 > 1$), so a cap set of size ≥ 2 has all $u_i \neq 0$; and two cap points never share a direction ($\langle \hat{u}, \hat{u}' \rangle = 1 > 1/5$). Thus in a cap set of size $N \geq 2$ the $\hat{u}_i$ are N distinct unit vectors in S^4 with pairwise inner products $\leq 1/5$.

Lemma 8 (LP bound, exact certificate). *Any N unit vectors in $\mathbb{R}^5$ with pairwise inner products $\leq 1/5$ satisfy $N \leq 16$.*

Proof. Let G_k denote the Gegenbauer polynomials for S^4 (parameter $\lambda = 3/2$) normalized by $G_k(1) = 1$, positive definite on S^4 (Schoenberg [9]): $G_1(t) = t$, $G_2(t) = (5t^2 - 1)/4$, $G_3(t) = (7t^3 - 3t)/4$. One has the exact identity

$$F(t) := 1 + (30/7)G_1(t) + (25/4)G_2(t) + (125/28)G_3(t) = (125/16)(t + 3/5)^2(t - 1/5),$$

with $f_0 = 1$, $f_1 = 30/7$, $f_2 = 25/4$, $f_3 = 125/28$, all coefficients positive, $F \leq 0$ on $[-1, 1/5]$, and $F(1) = 16$. For unit vectors $v_1, \dots, v_N$ with pairwise $t_{ij} \leq 1/5$, using $\sum_{i,j} G_k(t_{ij}) \geq 0$ (Schoenberg [9]),

$$N^2 = \sum_{i,j} G_0(t_{ij}) \leq \sum_k f_k \sum_{i,j} G_k(t_{ij}) = N F(1) + \sum_{i \neq j} F(t_{ij}) \leq 16N. \quad \square$$

Remark 9 (Sharpness). The bound is attained by the 5-demihypercube: the 16 vectors $(\pm 1, \dots, \pm 1)/\sqrt{5}$ with an even number of minus signs, whose pairwise inner products lie in $\{1/5, -3/5\}$. This is exactly half of the deep-hole family of D_5 — whence $M(D_5) = 16$ and the classical E_6 extension. The linear program alone therefore cannot give 15: the hole structure of Q_5, R_5 in the next lemma is essential.

Lemma 10 (Equality-case rigidity). $M(Q_5) \leq 15$ and $M(R_5) \leq 15$.

Proof. Suppose a cap set over $X \in \{Q_5, R_5\}$ had 16 points. Then equality holds throughout the proof of Lemma 8 for the $\hat{u}_i$ (the equality analysis of Delsarte–Goethals–Seidel [10]). In particular $\sum_{i \neq j} F(t_{ij}) = 0$ with every term ≤ 0 , so every t_{ij} lies in the root set $\{-3/5, 1/5\}$; and $\sum_{i,j} G_k(t_{ij}) = 0$ for $k = 1, 2, 3$ (all $f_k > 0$). For $k = 1$ this reads $0 = \sum_{i,j} G_1(t_{ij}) = \sum_{i,j} \langle \hat{u}_i, \hat{u}_j \rangle = |\sum_i \hat{u}_i|^2$, so $\sum_i \hat{u}_i = 0$, and pairing with $\hat{u}_i$: $1 + \sum_{j \neq i} t_{ij} = 0$. If n_i pairs equal $1/5$ (and $15 - n_i$ equal $-3/5$), then $1 + n_i/5 - 3(15 - n_i)/5 = 0$, i.e. $n_i = 10$ for every i : each point has ten partners at inner product exactly $1/5$.

By the strictness in Lemma 7, each such pair forces both endpoints to have $|u|^2 = 5/4$ exactly; since $n_i > 0$ for all i , all sixteen u_i have maximal norm, hence (Theorem 2, sphere-touching) all are deep-hole vertices, pairwise distinct. Their compatibility at $h_i = h_j = \sqrt{3}/2$ reads $\langle u_i, u_j \rangle \leq 1/4$: a 16-clique in the hole graph, contradicting Lemma 4. $\square$

Proof of Theorem 3. By Lemma 5, $|Y| = 40 + |Y^+| + |Y^-| \leq 40 + 15 + 15 = 70 < 72$. $\square$

Remark 11 (Calibration). Applied to D_5 or L_5 , the same chain yields no contradiction — their hole graphs do contain 16-cliques — and Lemmas 5–8 give $|Y| \leq 72$, exactly attained by the E_6 and Leech six-dimensional configurations. The argument is tight where it must be.

Conjecture 12. $M(Q_5) = M(R_5) = 12$; equivalently, the 64-point configurations of Theorem 6 are the largest six-dimensional kissing configurations with Q_5 or R_5 cross sections. (Supported by the exact hole-clique number 12 and by a multistart penalty minimization, 200 restarts per case, script `experiments/conjecture31_stageA.py`, which reaches 12 points to machine precision and finds no 13-point cap set; this probe is not part of the verified claims. Closing 12 versus 15 requires more than the LP bound, by Remark 9.)

5 Methodology, verification, and data

All verdict-relevant computations are exact: configurations are stored with exact coordinates (rational; the 64-point extensions carry $\sqrt{3}/2$ symbolically); an exact-arithmetic checker (canonical multiquadratic representation with a decidable sign test, no floating point on any accepting path) verifies all norm and inner-product claims; the vertex enumeration of Section 3 runs in integer arithmetic with proven overflow bounds; stresses carry rational certificates; ranks are recomputed exactly by two independent implementations and clique numbers by two algorithms. Three of the computational claims are checked by two independent computations: the jamming certificates of Theorem 1 (positivity and parallelism of the stress and the rank 190; one implementation uses dense integer rows and a λ -multiple test, the other sparse rows assembled from the contact graph, a 2×2 -minor test, different primes and an exact Bareiss rank), the vertex enumeration of Theorem 2 (a float-prefiltered enumeration with exact re-verification, and a tolerance-free integer enumeration), and the clique numbers of Lemma 4 (a library Bron–Kerbosch enumeration and an own branch-and-bound). The Gegenbauer identity

of Lemma 8 is checked by a single symbolic expansion; the 64-point configurations of Theorem 6 by the exact checker on the shipped coordinates and, separately, by rebuilding them from a 12-clique of the hole graph. Lemmas 5, 7 and 10 are proved in the text and use only the verified inputs above. The repository <https://github.com/smitherer/kissingnumbers> (cite as github.com/smitherer/kissingnumbers, release v1.0) contains all configurations, certificates, both implementations, and an experiment log; a single command re-runs every check against a table of expected values.

References

- [1] A. Korkine, G. Zolotareff, Sur les formes quadratiques, *Math. Ann.* **6** (1873), 366–389.
- [2] J. Leech, Five dimensional non-lattice sphere packings, *Canad. Math. Bull.* **10** (1967), 387–393.
- [3] J.H. Conway, N.J.A. Sloane, What are all the best sphere packings in low dimensions?, *Discrete Comput. Geom.* **13** (1995), 383–403.
- [4] H. Cohn, Y. Jiao, A. Kumar, S. Torquato, Rigidity of spherical codes, *Geom. Topol.* **15** (2011), 2235–2273.
- [5] H.D. Mittelmann, F. Vallentin, High-accuracy semidefinite programming bounds for kissing numbers, *Exp. Math.* **19** (2010), 175–179.
- [6] F. Szöllősi, A note on five dimensional kissing arrangements, *Math. Res. Lett.* **30** (2023), no. 5, 1609–1615; arXiv:2301.08272.
- [7] H. Cohn, I. Rajagopal, Variations on five-dimensional sphere packings, *Discrete Comput. Geom.* (2026), doi:10.1007/s00454-026-00841-x; arXiv:2412.00937.
- [8] D. de Laat, N. Leijenhorst, W.H.H. de Muinck Keizer, Optimality and uniqueness of the D_4 root system, arXiv:2404.18794.
- [9] I.J. Schoenberg, Positive definite functions on spheres, *Duke Math. J.* **9** (1942), 96–108.
- [10] P. Delsarte, J.M. Goethals, J.J. Seidel, Spherical codes and designs, *Geom. Dedicata* **6** (1977), 363–388.